Problem Sheet 6

1.

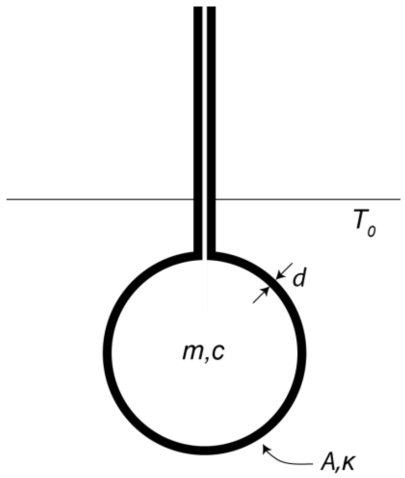
(a) A phase detector is fed with an input signal $V_{in} = A\sin(\omega_{in}t + \phi)$ and a reference signal in the form of a square wave at ω_{ref}. The phase difference between the two signals at $t = 0$ is ϕ. The output is averaged over one cycle of ω_{ref}. For the case $\omega_{ref} = \omega_{in} = \omega$, write the integral form of the average $\langle V_{out} \rangle$.

(b) Show that

$$\langle V_{out} \rangle = \frac{2A\cos\phi}{\pi}.$$

2. SEMINAR QUESTION

(a) The figure below shows a thermometer made from a mercury-filled glass bulb. The liquid may expand into the tube to indicate the temperature and we will assume that the tube is thin and that we can ignore its effect. The thermometer has an initial temperature of 0°C and is plunged into a heat-bath of temperature T_0 at time $t = 0$. The glass bulb has thickness d, surface area A and thermal conductivity κ (units W/K/m). The mercury inside the bulb has mass m, specific heat capacity c and an extremely high thermal conductivity such that it can be considered isothermal (for a traditional mercury thermometer this is a reasonable assumption).



If T is the temperature in °C at time $t > 0$, write an expression for the rate of heat transfer dQ/dt where Q is the heat (energy) crossing the glass barrier from the heat-bath to the mercury.

Hint: use dimensional analysis to derive the relationship between Q, t, κ, ΔT (the temperature difference between the mercury and the heat bath) and the dimensions of the glass bulb A, d.

(b) Derive an expression for the temperature T for time $t > 0$.

(c) Considering the heat Q to be equivalent to electrical charge q, what quantities in this thermal system can be said to be equivalent to

1. Electrical resistance
2. Capacitance
3. Voltage

Which electrical circuit exhibits a similar behaviour to the thermometer?